open data | data on request | closed data | preprint | reanalysis on open data | our reproduction

(a) Maximum vocabulary size used in the 32 English communication-neuroprostheses studies

vocabulary size

71,400 word forms (Brysbaert et al., 2016)
42,000 lemmas (Brysbaert et al., 2016)
11,100 word families (Brysbaert et al., 2016)

6 | 8 | 5–10 | 50 | 178 | 1,024 | 1,024 | 1,152 | 100k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 125k | 130k | 130k | open | open | open | open

Angrick et al., 2024 (CC01) | Wandelt et al., 2024 (P1/P2) | Brosler et al., 2026 (B1r/B3/B6) | Moses et al., 2021 (B1) | Silva et al., 2024b (B1) | Metzger et al., 2023 (B3) | Littlejohn et al., 2025 (B3) | Metzger et al., 2022 (B1) | Feghhi et al., 2026 (T12/T15) | Willett et al., 2023 (T12) | Card et al., 2024 (T15) | Feng et al., 2024 (T12) | Kunz et al., 2025 (T12/T15/T16) | Li et al., 2025 (T12) | Feghhi et al., 2025 (T12) | Levin et al., 2026 (T12/T15/T16) | Y. Zhang et al., 2026 (T12/T15) | Nason-Tomaszewski et al., 2026 (T16) | Boccato et al., 2026 (T12/T15/T16/T17) | Jude et al., 2026a (T17/T18) | Jude et al., 2026b (T17) | Yoon et al., 2026 (T12) | Card et al., 2026 (T15) | Fogg et al., 2026 (T12/T15/T16/T17/T21/T22) | Z. Zhang et al., 2026 (T12/T15) | Pouratian et al., 2026 (V1/V2/V3) | Fan et al., 2023 (T5) | Karpowicz et al., 2024 (T5) | Willett et al., 2021 (T5) | Wairagkar et al., 2025 (T15) | Levin et al., 2026 (T5/T12/T18) | Wairagkar et al., 2026 (T15)

Filtered to 8 open-data studies exceeding native-speaker vocabulary (>71,400 word forms)

(b) Open-sourced dataset size

released sentences

10 d/sess, 95 blks, span 253 d: 1,000
24 d/sess, 275 blks, span 120 d: 9,680
21 d/sess, 105 blks, span 287 d: 1,315
45 d/sess, 265 blks, span 612 d: 9,498
37 d/sess, blks n/a, span 171 d: 8,489
1 d/sess, 12 blks, span 1 d: 500
3 d/sess, 18 blks, span 309 d: 720
2 d/sess, 14 blks, span 234 d: 450
8 d/sess, 46 blks, span 89 d: 668
9 d/sess, 98 blks, span 119 d: 1,033
13 d/sess, 96 blks, span 141 d: 3,674

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | ★ Fan et al., 2023 (T5) | ★ Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | ★ Jude et al., 2026a (T17) | ★ Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

(c) Word error rate (online)
★ 3 studies satisfy: Open Data, Vocab > 71.4k, WER < 12.8%

† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days

word error rate (%)

12.8% 14-dataset average (Radford et al., 2022)
5.8% Switchboard (Xiong et al., 2017)
2.5% LibriSpeech (Radford et al., 2022)

vocab open mean±SD† 5 of 10 d: 25.1±8.6
vocab 125k mean±SD† 5 of 24 d: 23.9±1.8
vocab 130k mean±SD† 10 of 21 d: 4.0±1.2
vocab 125k mean±SD† 14 of 45 d: 4.9±3.3
vocab open listener median 8 of 37 d: 43.8
vocab 50-w mean±SD† 1 of 1 d: 23.7±0.0
vocab 125k mean±SD† 2 of 3 d: 51.3±33.1
vocab 125k mean±SD† 1 of 2 d: 46.6±0.0
vocab 125k mean±SD† 6 of 8 d: 12.5±2.5
vocab 125k mean±SD† 7 of 14 d: 3.4±0.8
vocab 125k mean±SD† 4 of 13 d: 66.5±18.1

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | ★ Fan et al., 2023 (T5) | ★ Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | ★ Jude et al., 2026a (T17) | ★ Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

(d) Communication speed (online)
n/r: speed not reported

† recomputed from released data (decode outputs / trial timing / Source Data files)
‡ digitized from the paper's figure markers
error bar is calculated across days

words per minute

160 wpm natural conversation (Card et al., 2024)
51.6 wpm keyboard typing (Dhakal et al., 2018)
36.2 wpm smartphone typing (Palin et al., 2019)
25.1 wpm handwriting (Kaifosh et al., 2025)
5–10 wpm eye typing (Hamid & Kristensson, 2024)
8 wpm point-and-click BCI (Willett et al., 2021)

vocab open mean±SD† 5 of 10 d: 15.4±2.2
vocab 125k mean±SD† 5 of 24 d: 65.3±0.9
vocab 130k mean±SD† 15 eval d (10 of 21 d): 14.6±2.2
vocab 125k mean±SD† 14 of 45 d: 32.5±2.7
vocab 125k mean±SD† 6 of 8 d: 6.8±1.8
vocab 125k mean±SD† 7 of 14 d: 14.0±4.5
vocab 125k mean±SD† 8 of 13 d: 34.1±7.7

Willett et al., 2021 (T5) | Willett et al., 2023 (T12) | ★ Fan et al., 2023 (T5) | ★ Card et al., 2024 (T15) | Wairagkar et al., 2025 (T15) | Kunz et al., 2025 (T12) | Kunz et al., 2025 (T15) | Kunz et al., 2025 (T16) | ★ Jude et al., 2026a (T17) | ★ Jude et al., 2026a (T18) | Jude et al., 2026b (T17)

Our reproduction on a 64 GB RAM / RTX 5060 Ti (16 GB) home PC

(e) Reproduction: acoustic / synthesis stage (no LM), one pair per panel-(c) bar

our reproduction (16 GB home PC) | paper / benchmark anchor
error bar is calculated across days

Greedy PER / CER / 1−r (%)

Willett et al., 2021 (T5, raw CER): 4.7±1.9 | 2.8±1.4
Willett et al., 2023 (T12, greedy PER): 23.4±3.2 | 19.7
Fan et al., 2023 (T5, CORP vo CER): seed: 2.65 | 6.2±3.3 | 6.2±2.3
Card et al., 2024 (T15, greedy PER): 10.2±4.2 | 9.5±2.8
Wairagkar et al., 2025 (T15, 1−mel r²%): r: 0.69 vs 0.83 | 31.4±5.2 | 17.0±4.0
Kunz et al., 2025 (T12, greedy PER): 50-w sess. | 61.4±0.0 | 24.0±0.0
Kunz et al., 2025 (T15, greedy PER): 3-sess. subset | 59.7±8.4 | 54.0±33.1
Kunz et al., 2025 (T16, greedy PER): 2-sess. subset | 61.5±12.1 | 52.0±0.0
Jude et al., 2026a (T17, raw CER): best 8.2% | 19.4±5.8 | 18.6±5.8
Jude et al., 2026a (T18, raw CER): best 2.6%; <12% | 7.7±2.8 | 12.0±2.4
Jude et al., 2026b (T17, greedy PER): 40.0±9.2 | 34.0±6.2

(f) Reproduction: full-pipeline WER, one pair per panel-(c) bar

our reproduction (16 GB home PC) | paper / benchmark anchor
error bar is calculated across days

Word error rate (%)

Willett et al., 2021 (T5): bigram 63-tr. pool | 10.0±3.6 | 3.6±3.2
Willett et al., 2023 (T12): day med. 5-gram | 23.6±5.2 | 23.8±4.0
Fan et al., 2023 (T5): recal. arm 10 sess. | 4.1±1.2 | 6.2±2.3
Card et al., 2024 (T15): 41-s pool biGRU-48k | 6.2±6.5 | 6.7±6.6
Wairagkar et al., 2025 (T15): WER n/r (voice; see e)
Kunz et al., 2025 (T12): 50-w sess. gap attr. | 94.3±0.0 | 24.0±0.0
Kunz et al., 2025 (T15): 125k gap attr. | 78.1±0.6 | 54.0±33.1
Kunz et al., 2025 (T16): 125k gap attr. | 82.4±0.0 | 52.0±0.0
Jude et al., 2026a (T17): best 4.6% | 11.4±6.8 | 12.5±2.5
Jude et al., 2026a (T18): best 0.0%; <7% | 6.7±3.7 | 7.0±2.4
Jude et al., 2026b (T17): conv. day-68 | 44.4±16.0 | 50.0±18.1

(g) Host RAM across all 8 studies: our 64 GB PC vs. official stacks

our reproduction (64 GB RAM PC) | official / naive stack | 64 GB consumer PC RAM ceiling

Peak host RAM (GB)

Willett et al., 2021 (T5): mmap + lazy FST | 26.9 | 56
Willett et al., 2023 (T12): mmap + .idx | 19.6 | 300
Fan et al., 2023 (T5): TE GC fix | 18.4 | 40
Card et al., 2024 (T15): mmap + .idx | 19.6 | 300
Wairagkar et al., 2025 (T15): stream win. | 1.8 | 24
Kunz et al., 2025 (T12/15/16): mmap + .idx | 19.6 | 300
Jude et al., 2026a (T17/T18): drop unpruned FST | 22.5 | 64
Jude et al., 2026b (T17): mmap + .idx | 19.6 | 300

(h) GPU VRAM across all 8 studies: our 16 GB GPU vs. official rigs

our peak VRAM (1× RTX 5060 Ti) | official provisioned GPU VRAM | 16 GB single-GPU VRAM ceiling

Peak GPU VRAM (GB)

Willett et al., 2021 (T5): RNN + GPT-2 | 4 | 44
Willett et al., 2023 (T12): FP16 + OPT | 13 | 80
Fan et al., 2023 (T5): seed + CORP | 4 | 40
Card et al., 2024 (T15): accum + OPT | 13 | 48
Wairagkar et al., 2025 (T15): causal Mel | 1.5 | 72
Kunz et al., 2025 (T12/15/16): FP16 + OPT | 13 | 48
Jude et al., 2026a (T17/T18): BF16 port | 11.5 | 24
Jude et al., 2026b (T17): FP16 + OPT | 13 | 48

(i) Session-level PER vs. post-LM WER

word error rate after LM stage (%)
greedy phoneme error rate per session (%)

No-LM baseline: 1 − (1 − PER)^2.4
WER = PER
PER = 25%
(≤25%: 45/57 = 79% WER < PER;
>25%: 4/20 = 20% WER < PER)
This means that we must get PER < 25%.

per-session (open) | pooled (star)
Willett et al., 2023 (T12)
Card et al., 2024 (T15)
Jude et al., 2026b (T17)